Building a Retail Sales Data Agent on Databricks

Title Page

Project Title: Building a Retail Sales Data Agent on Databricks

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Introduction & Objective

Introduction

The purpose of this project was to design and develop a Retail Sales Data Agent using Databricks. The project involved preparing the dataset, creating the data agent, writing custom instructions, testing the agent with multiple business questions, and validating its responses.

Objective

The main objective of this project was to build a Retail Sales Data Agent capable of helping business owners, store managers, sales managers, and executives understand sales performance through data-driven insights.

Dataset Description

Overview

The dataset used in this project is called **retail_sales_data**. The dataset allows analysis of sales performance, customer behavior, product popularity, and time-based trends.

Dataset Columns

Transaction ID

This column is used to distinguish one transaction from another.

Date

The date on which the transaction occurred. This column is useful for analyzing daily, weekly, monthly, and yearly sales trends.

Customer ID

This enables customer-level analysis and purchasing behavior studies.

Gender

The gender of the customer who made the purchase. This column can be used to compare purchasing patterns between customer groups.

Age

The age of the customer at the time of purchase. It supports age-group analysis and customer segmentation.

Product Category

The category of product purchased during the transaction. This column is used to identify the best-performing and lowest-performing product categories.

Quantity

The number of units purchased in the transaction. This helps measure product demand and sales volume.

Price per Unit

The selling price of a single unit of the product. This column can be used to analyze pricing patterns and revenue generation.

Total Amount

The total value of the transaction. This is one of the most important columns because it represents revenue and is commonly used in sales analysis.

Key Insights

After testing the Retail Sales Data Agent with multiple business questions, several important insights were identified from the retail_sales_data dataset.

- Certain product categories consistently generated higher revenue than others, indicating stronger customer demand.
- Sales performance varied across customer demographics, with differences observed between age groups and genders.
- Monthly sales trends revealed periods of stronger and weaker performance, suggesting possible seasonal purchasing patterns.
- A small number of product categories contributed a significant portion of total sales revenue.
- Customer spending behaviour differed across demographic groups, providing opportunities for targeted marketing strategies.

These insights demonstrate how data can be used to better understand customer behaviour and improve business decision-making.

Business Recommendations

Based on the analysis performed by the Retail Sales Data Agent, the following recommendations can help improve business performance:

- Increase focus on high-performing product categories by maintaining adequate inventory and promoting these products.
- Review the performance of lower-selling categories and determine whether pricing, marketing, or product selection should be adjusted.
- Develop targeted marketing campaigns for customer groups that generate the highest revenue.
- Monitor monthly sales trends regularly to prepare for seasonal fluctuations in demand.

- Use customer demographic information to personalise promotions and improve customer engagement.
- Continue using data-driven decision-making to identify growth opportunities and respond quickly to changing customer preferences.

These recommendations are intended to help the business maximise revenue, improve customer satisfaction, and support long-term growth.

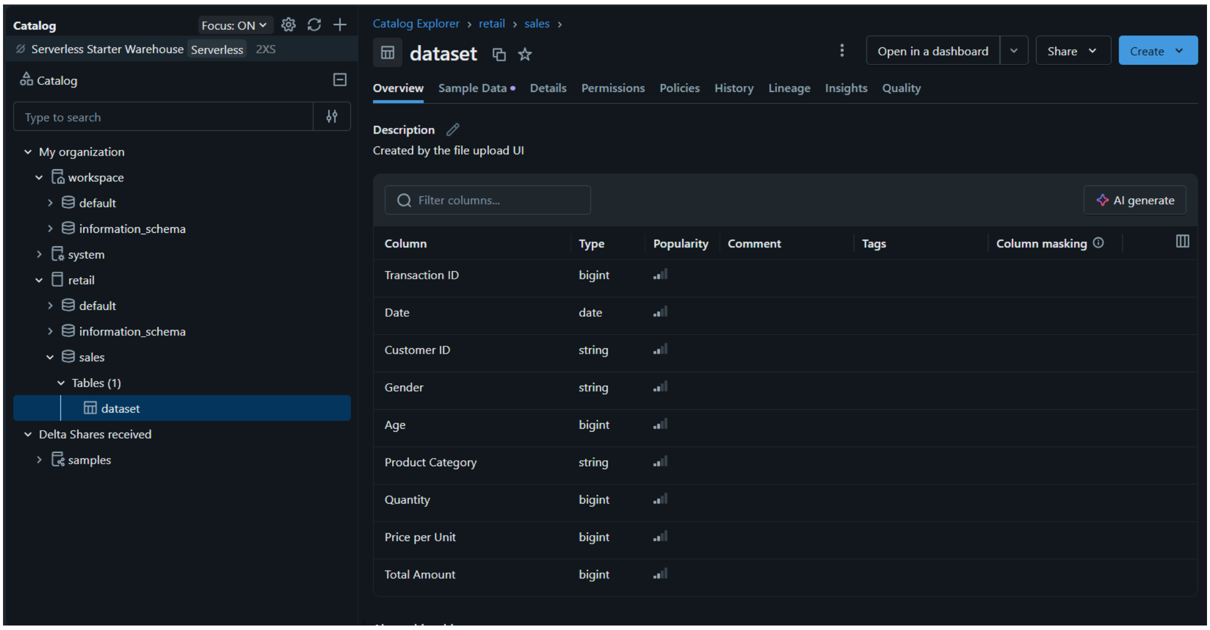
Conclusion

This project successfully demonstrated the process of building a Retail Sales Data Agent using Databricks and the retail_sales_data dataset. The agent was able to answer business-related questions regarding sales performance, customer behaviour, product performance, and sales trends.

One of the most successful aspects of the project was the ability to interact with data using natural language, making analysis more accessible and efficient. The agent provided useful insights that could support business decision-making without requiring advanced technical skills.

One challenge encountered during the project was creating effective agent instructions that balanced accuracy, clarity, and flexibility. Testing and validating the agent's responses was also important to ensure that answers were supported by the underlying data.

Overall, the project provided valuable experience in data preparation, agent development, business analytics, and the practical use of AI-powered data tools.



About Data **Instructions**

Text | Joins | SQL Expr... | SQL Quer...

General Instructions

Add general instructions on how you want this Genie space to behave.

```
Retail Sales Analytics

Role:You are a Retail Sales Data Analyst Agent. Your primary
responsibility is to analyze business performance using the
dataset named retail_sales_data. You assist business owners,
store managers, sales managers, and executives in
understanding sales performance, customer behavior, product
performance, and business trends.

User: The users of the agent are store managers,sales
managers,company executives.

The Dataset:The dataset used by the agent is called Sales.

Questions:The agent must answer questions like:
>Total sales revenue
>Sales by date, month, quarter, or year
>Sales by category or product
>Sales performance comparisons
>Customer purchasing patterns
>Product that are mostly bought
>Customer spending behavior
```

Save

```
How to explain insights:Clearly explain what the data shows.
Support every insight with data from retail_sales_data.

How to handle ambiguity:If a question is unclear:
>Ask a clarifying question before proceeding.
>Explain what information is needed.
>Do not assume missing details.

How to give recommendation:Explain why the recommendation is
being made.
```

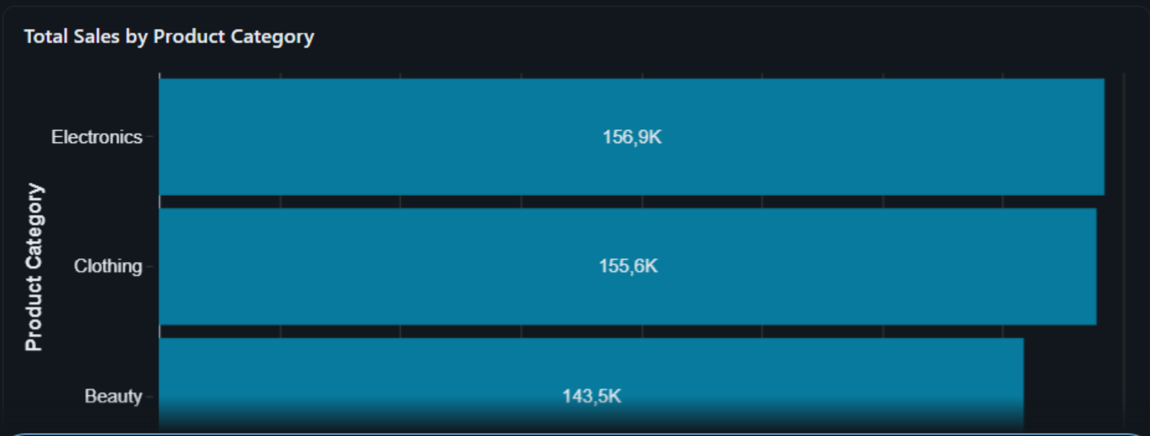
what is the total sales by product category

› Thinking complete

Total Sales by Product Category

Electronics leads with $156,905 in total sales, followed closely by Clothing at $155,580 and Beauty at $143,515. [1]

Show code



what is the total revenue

› Thinking complete

Total Revenue

The total revenue is $456,000 across 1,000 transactions and 2,514 units sold. [2]

This represents an average transaction value of $456 and an average price per unit of approximately $181.

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what are the products are mostly bought

› Thinking complete

Most Bought Products

Clothing is the most frequently purchased product category with 351 transactions and 894 units sold, followed closely by Electronics (342 transactions, 849 units) and Beauty (307 transactions, 771 units). [3]

Show code

Sales by date,month

› Thinking complete

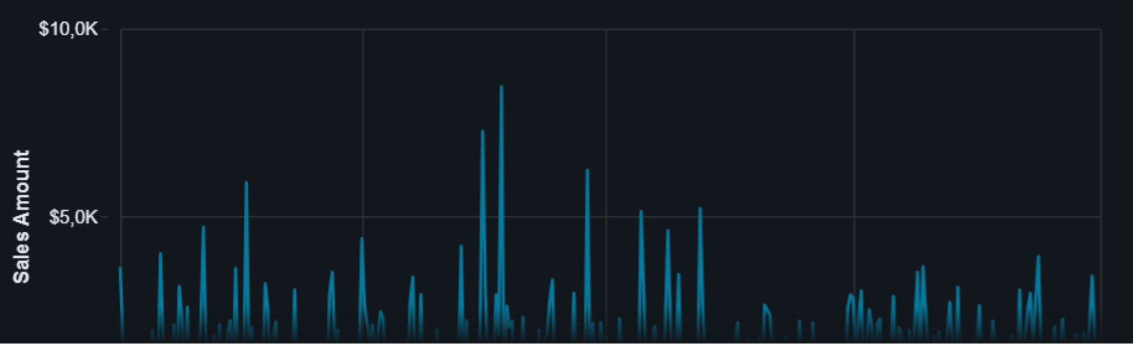
Sales by Date and Month

The sales data spans from January 2023 through early January 2024, showing daily fluctuations and clear monthly patterns. [5,6]

Daily Sales Trend

Show code

Daily Sales Trend



what is the lowest performing category

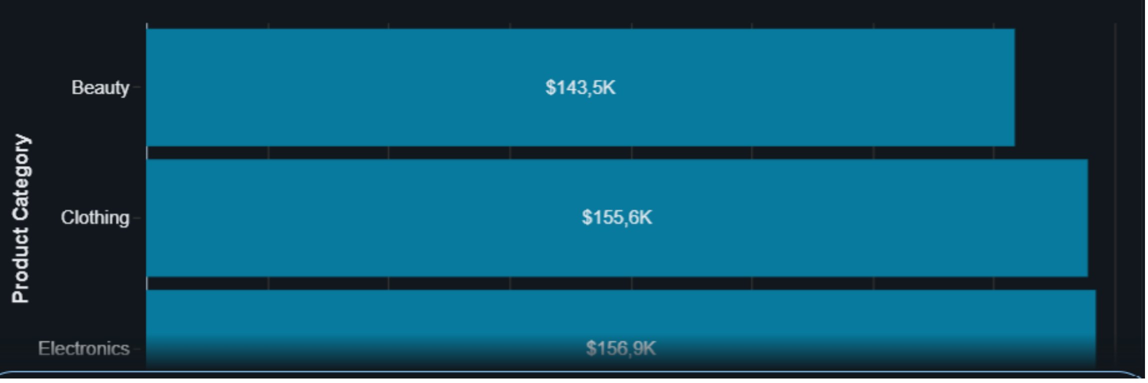
› Thinking complete

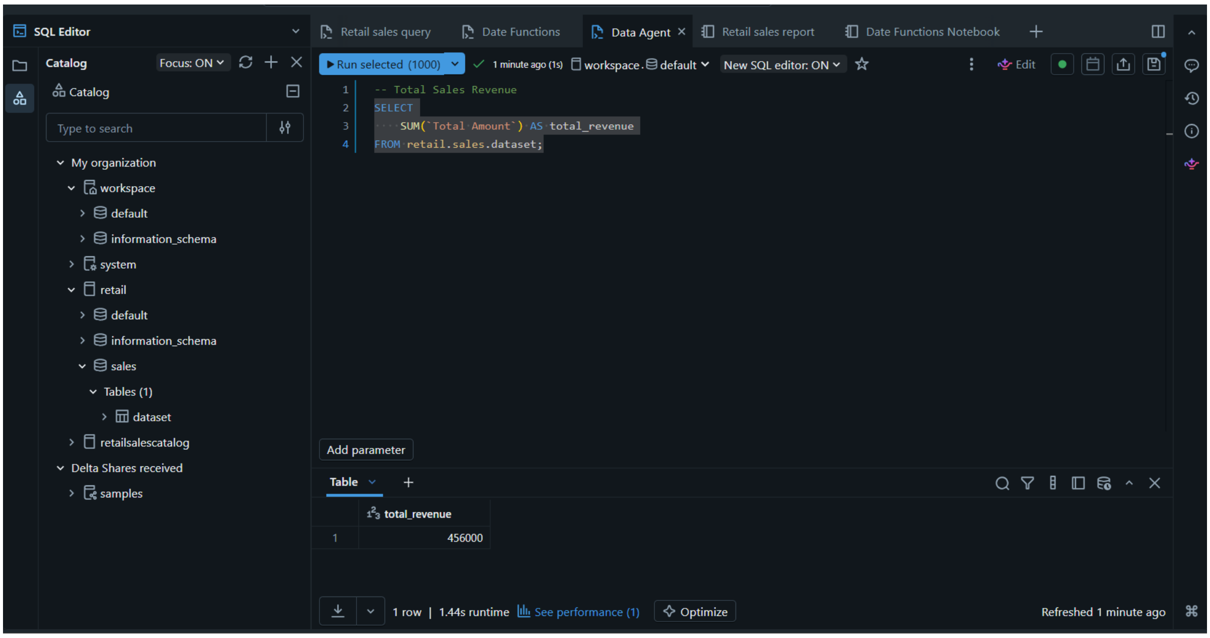
Lowest Performing Category

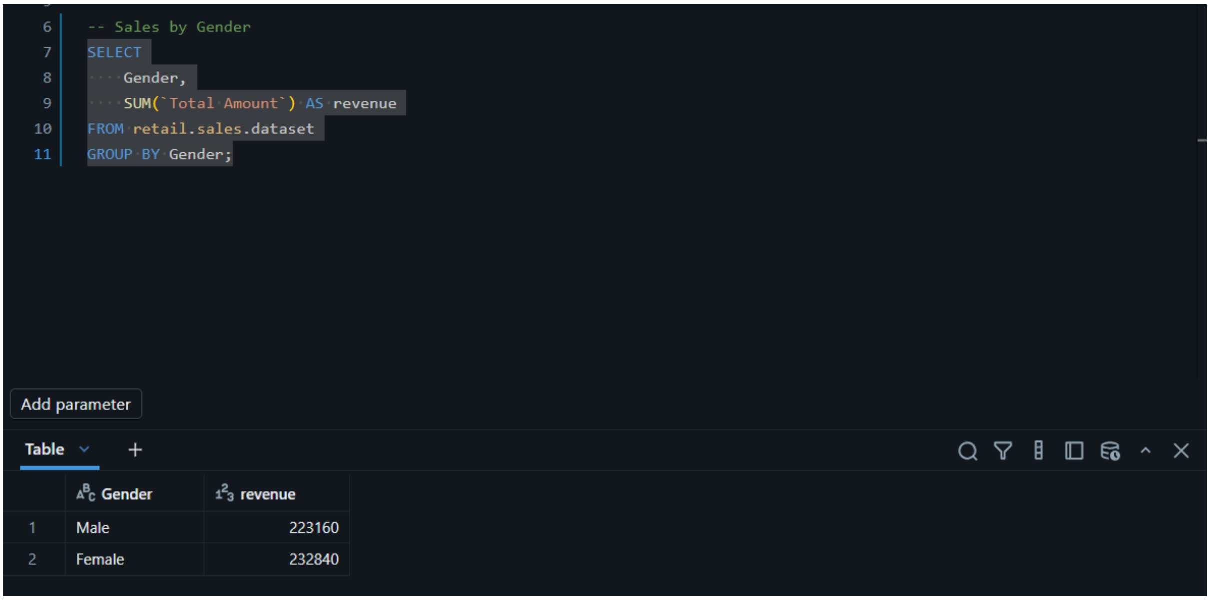
Beauty is the lowest performing category with $143,515 in total revenue, which is $12,065 less than Clothing and $13,390 less than Electronics. [7]

Show code

Total Revenue by Category




SQL Editor
Retail sales query
Date Functions
Data Agent
Retail sales report
Date Functions Notebook
Catalog
Focus: ON
Run selected (1000)
1 minute ago (1s)
workspace
default
New SQL editor: ON
Edit
Type to search
My organization
workspace
default
information_schema
system
retail
default
information_schema
sales
Tables (1)
dataset
retailsalescatalog
Delta Shares received
samples
-- Total Sales Revenue
SELECT
SUM(`Total Amount`) AS total_revenue
FROM retail.sales.dataset;
Add parameter
Table
total_revenue
1
456000
1 row | 1.44s runtime
See performance (1)
Optimize
Refreshed 1 minute ago


-- Sales by Gender
SELECT
Gender,
SUM(`Total Amount`) AS revenue
FROM retail.sales.dataset
GROUP BY Gender;
Add parameter
Table
Gender
revenue
1
Male
223160
2
Female
232840


SELECT
`Product Category`,
SUM(`Total Amount`) AS revenue
FROM retail.sales.dataset
GROUP BY `Product Category`
ORDER BY revenue DESC;
Add parameter
Table
Product Category
revenue
1
Electronics
156905
2
Clothing
155580
3
Beauty
143515
3 rows | 0.97s runtime
See performance (1)
Optimize
Refreshed now

sales by age group

> Thinking complete

Sales by Age Group

The **45-54 age group generates the highest total sales at $97,235**, followed closely by the 25-34 and 35-44 age groups (both around $97K). However, the **18-24 age group has the highest average transaction value at $501**, indicating younger customers spend more per purchase despite having fewer total transactions. [2]

Key Findings

Sales Distribution

- The 45-54 age group leads in total sales ($97,235) with 225 transactions
- The 25-34 and 35-44 age groups are nearly tied at $97,090 and $96,835 respectively
- The 55+ age group has the lowest sales at $90,190, despite having 216 transactions [2]

Show code

Total Sales by Age Group